Aristotle's Pyramid

Theorem: of the five face pyramids of the Platonic solids, only the cube's alone and the tetrahedron's with the octahedron's fill space; the icosahedron's, the Cleo-Icos, needs a partner that is not a face pyramid

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Version 3 · 3 November 2026 (Version 2: 22 October 2026; Version 1: 21 October 2026) · DICTO DOI: 10.5281/zenodo.23529497 (concept) · View in app: https://kaleidohedra.dictospheres.com/?open=aristotle:pyramids · Licence CC BY-SA 4.0 · Scripts, exact checks and figures: https://github.com/DICTOR-Master/kaleidohedra/tree/research-fig3/studies/2026-10-21-aristotle-pyramids (research branch, commit research-fig3; also attached to this record)
The idea that Aristotle may have meant the Cleo-Icos is DICTO's, as are the names Cleo-Icos, Antonius and Admiral Caps and the earlier findings of §5; computation, proofs and drafting with Claude (Anthropic).
Version 2 adds a short opening explanation (in italics) to each result, its proof and the conclusion; the mathematics is unchanged.
Version 3 adds the series line: this paper is Part I of *The Centre Pyramids: from Plato to Kepler*; Part II, DICTO Vacuus Solids (paper #87, DICTO DOI: 10.5281/zenodo.23896016), studies the walls between the same centre pyramids, their turning out into the Star, and DICTO's slotted cardboard builds; DICTO's part is now written as what he proposed. The mathematics is unchanged.

Abstract. Main theorem. Of the five face pyramids of the Platonic solids (the cone from the centre over one face), a set fills space exactly when it is the cube's pyramid alone, the tetrahedron's and the octahedron's pyramids together, or the union of these two; no other set fills space. Here a set fills space when copies of its pyramids, in any numbers, any positions and mirror images, every kind used and not necessarily face to face, tile space. The proof is a Dehn-type counting map on dihedral angles, made rigorous with the Splitting Theorem of Conway, Radin and Sadun: the dodecahedron's and the icosahedron's pyramids can never occur, and the tetrahedron's occurs exactly when the octahedron's does; the octet truss cut from its centres and the cubic tiling give the tilings. Corollaries: alone only the cube's pyramid fills space, in pairs only the tetrahedron's with the octahedron's, and with equal edges their numbers are asymptotically equal; the icosahedron's pyramid, the Cleo-Icos (DICTO's name), fills space only with a partner that is not a face pyramid, Antonius (#20). The question is DICTO's: Aristotle wrote that among solids only "the pyramid and the cube" fill space (On the Heavens III.8); read as the regular tetrahedron this is false, and DICTO proposed that he may have meant the Cleo-Icos. A section after the theorem collects DICTO's earlier exact fillings by the whole Platonic solids with partners. Every angle identity, the test over all 31 sets and the constructions are checked in exact arithmetic.

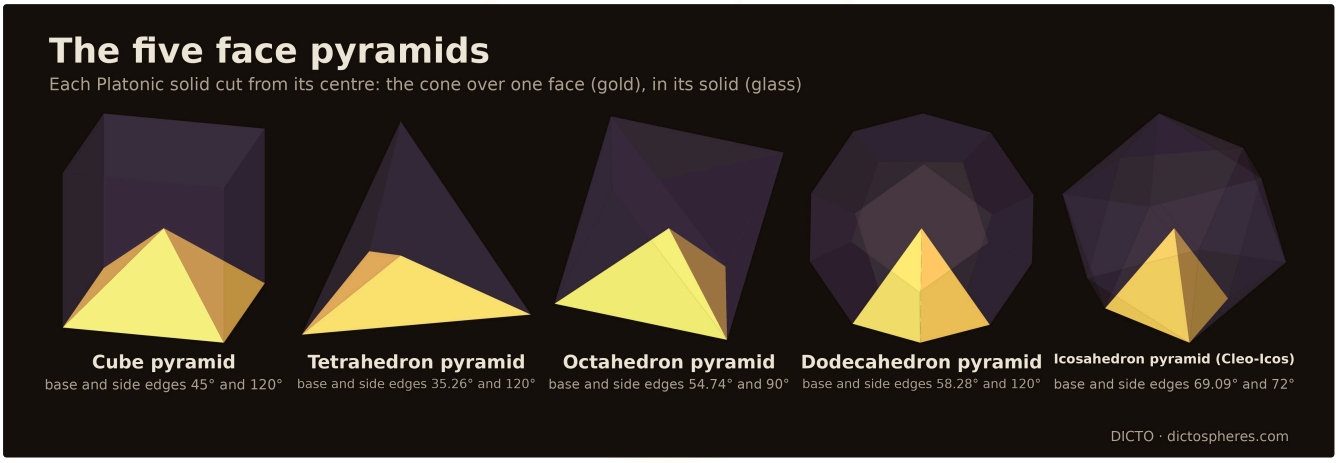


Figure 1. The five face pyramids (gold), each shown in its solid (glass) with its tip at the solid's centre and the face at the bottom: cube, tetrahedron, octahedron, dodecahedron and icosahedron (the Cleo-Icos). Under each: the dihedral angles along its base edges and its side edges (Table 1).

1. The theorem

Face pyramids. For a Platonic solid with centre O, the face pyramid over a face F is the convex hull of F and O (Figure 1). Its dihedral angle along a base edge is half the solid's dihedral angle, since the plane through O and the edge bisects it; along a side edge (from O to a corner) it is 360°/q, where q faces meet at that corner. We write C, T, O, D and I for the face pyramids of the cube, tetrahedron, octahedron, dodecahedron and icosahedron; I is DICTO's Cleo-Icos, reached from his CLEO piece [2, 3].

Face pyramid	Base edges	Dihedral along a base edge	Side edges	Dihedral along a side edge
cube	4	45° = π/4	4	120° (3 faces at a corner)
tetrahedron	3	α/2 ≈ 35.26°	3	120°
octahedron	3	(π − α)/2 ≈ 54.74°	3	90° (4 faces)
dodecahedron	5	$\theta_D/2 \approx 58.28°$	5	120°
icosahedron (Cleo-Icos)	3	$\theta_I/2 \approx 69.09°$	3	72° (5 faces)

Table 1. The five face pyramids. $\alpha = \arccos(1/3) \approx 70.53°$, $\theta_D = \arccos(-1/\sqrt{5})$ and $\theta_I = \arccos(-\sqrt{5}/3)$ are the dihedral angles of the tetrahedron, dodecahedron and icosahedron; the octahedron's is π − α (its cosine is −1/3) and the cube's π/2.

Filling space. A set U of face pyramids *fills space* if there is a tiling of space whose tiles are congruent copies (mirror images allowed) of pyramids of U, at finitely many fixed sizes, in any numbers and any positions, every kind in U used, with no face-to-face condition.

> *Cut each Platonic solid from its centre into identical pyramids, one on every face. Aristotle wrote that the pyramid fills space; DICTO proposed that he may have meant one of these. The theorem gives the complete answer: only the cube's pyramids fill space on their own, and only the tetrahedron's and the octahedron's do it together. The dodecahedron's and the icosahedron's can never be used at all.*

> **Main theorem.** A non-empty set U of the five face pyramids fills space if and only if U is {C}, {T, O} or {C, T, O}: the cube's pyramid alone, the tetrahedron's and the octahedron's pyramids together, or both of these. More precisely, in every tiling of space by copies of face pyramids, (a) no tile is a D or an I, and (b) T tiles occur if and only if O tiles occur.

2. Proof

> *The proof rests on a simple fact: wherever pieces meet along an edge, their angles round that edge must add up to a full turn.*

Lemma 1 (angles round an edge). Let space be tiled by convex polyhedra, finitely many meeting any bounded set, and let e be an edge of a tile P. For all but finitely many points x of e, the tiles containing x have x either in the relative interior of an edge on the line of e, or in the relative interior of a face whose plane contains that line; the wedge angles at x (the dihedral angles of the tiles with x on an edge) add up to 2π, or to π when one tile has x inside a face (the edge lies on a face).

Proof. Only finitely many tiles meet a neighbourhood of e. Leave out of e the finitely many points that are corners of these tiles, or where an edge of one of them crosses the line of e. At any other point x of e, no tile has x in its interior (it would overlap P), and a tile with x inside an edge has that edge on the line ℓ of e. If a tile T has x inside a face, near x it is a half-space; P lies in the other closed half-space, and so does the line ℓ through x, so the plane contains ℓ. Near x every tile is therefore a wedge or a half-space bounded by planes through ℓ, and in the plane at right angles to ℓ through x they become sectors that cover a disc with disjoint interiors: their angles add up to 2π. A half-space counts π, and there is at most one, since two would leave no room for P. ∎

Lemma 1 is the edge-counting step behind the classical fact that a polyhedron which tiles space has Dehn invariant zero (Debrunner [9]; Lagarias and Moews [10]); we prove it here directly because the tilings may use several kinds of tiles.

Now let h be a map from angles to the rationals that is additive with rational coefficients and has $h(\pi) = 0$. At every such point x the values of h at the dihedral angles there add up to $h(2\pi) = 0$ or $h(\pi) = 0$. Such maps exist with any values on α, θ_D and θ_I, because:

> *The awkward angles of the tetrahedron, dodecahedron and icosahedron are unrelated to one another, so none of them can make up for another.*

Lemma 2. π, α, θ_D and θ_I are linearly independent over the rationals.

Proof. $\cos^2\alpha = 1/9$, $\cos^2\theta_D = 1/5$ and $\cos^2\theta_I = 5/9$, so $\tan\alpha = 2\sqrt{2}$, $\tan\theta_D = -2$ and $\tan\theta_I = -2/\sqrt{5}$: all three are pure geodetic angles in the sense of Conway, Radin and Sadun [4], with tangents rational multiples of $\sqrt{2}$, $\sqrt{1}$ and $\sqrt{5}$, three different square classes. None is a rational multiple of π: if $\theta = (m/n)\pi$ then $e^{i\theta}$ is a root of unity and $2\cos\theta$ an algebraic integer, but $(2\cos\theta)^2$ is 4/9, 4/5 and 20/9, not integers. (Theorem 2 of [4] writes each angle as a rational multiple of π plus their basis angles; this shows the basis part is not zero.) Suppose a rational combination $q_1\alpha + q_2\theta_D + q_3\theta_I$ is a rational multiple of π. By the Splitting Theorem of [4, Theorem 3], so is each part whose tangents are rational multiples of one square root: $q_1\alpha$, $q_2\theta_D$ and $q_3\theta_I$ separately; hence $q_1 = q_2 = q_3 = 0$. ∎

In this basis the cube pyramid's angles are $\pi/4$ and $2\pi/3$, the side edges' $2\pi/3$, $\pi/2$ and $2\pi/5$, all rational multiples of π ($h = 0$); the base edges carry $\frac{1}{2}\alpha$ (T), $\frac{1}{2}\pi - \frac{1}{2}\alpha$ (O), $\frac{1}{2}\theta_D$ (D) and $\frac{1}{2}\theta_I$ (I). This is the argument of #20 for the Cleo-Icos alone [3, §5.6–5.7], carried out for all five and made rigorous with [4].

> *Keep count of one awkward angle at a time. A dodecahedron or icosahedron pyramid leaves an angle nothing can complete; a tetrahedron pyramid's angle can be completed only by an octahedron pyramid's, and the other way round.*

Proof of (a) and (b). (a) Choose $h(\theta_D) = 1$ and $h(\pi) = h(\alpha) = h(\theta_I) = 0$. The base edges of a D carry $h = \frac{1}{2}$, and every other dihedral angle of every pyramid carries 0. At a point x of a base edge of a D, chosen as in Lemma 1, the values of h add up to at least $\frac{1}{2}$, not 0: a contradiction. The same with $h(\theta_I) = 1$ excludes I. (b) Choose $h(\alpha) = 1$, the rest 0: a T's base edges carry $+\frac{1}{2}$, an O's $-\frac{1}{2}$, all other angles 0. Without O, a base edge of a T gives a sum of at least $\frac{1}{2}$; without T, a base edge of an O gives at most $-\frac{1}{2}$. So a set that fills space contains neither D nor I, and contains T exactly when it contains O: it is {C}, {T, O} or {C, T, O}.

> *And these three ways really work: six cube pyramids make a cube, and the classical octet of tetrahedra and octahedra, cut from their centres, gives the other two.*

The three sets fill space. {C}: six cube pyramids make a cube. {T, O}: Maurolico's octet [1] of regular tetrahedra and octahedra, each cut from its centre into its face pyramids. Two tetrahedra on opposite faces of an octahedron make a rhombohedron of 60°, one cell of the face-centred cubic lattice [1]; with lattice vectors (1, 1, 0), (1, 0, 1), (0, 1, 1) the octahedron is 4/3, each tetrahedron 1/3 and the cell 2, and it cuts into 8 octahedron pyramids and 8 tetrahedron pyramids (Figure 2, top). Round each edge of the octet, between 2 tetrahedra and 2 octahedra, the pyramids give $4 \times \alpha/2 + 4 \times (\pi - \alpha)/2 = 2\pi$. {C, T, O}: place the octet with its octahedra centred at the integer points of odd coordinate sum (corners at distance 1 along the axes) and its tetrahedra in the unit cubes; then every face pyramid lies in a slab $k \le z \le k + 1$, so the plane $z = 0$ is made of pyramid faces. Keep the octet's pyramids below it and fill the half-space above it with unit cubes cut into cube pyramids. ∎

Verification. pyramid_checks.py checks in exact arithmetic every angle in Table 1 (through $\cos^2$ and the sign of cos), the facts used in Lemma 2, the test (a)–(b) over all 31 non-empty sets of the five pyramids (only {cube}, {tetrahedron, octahedron} and {cube, tetrahedron, octahedron} pass, and with all five allowed D and I are still excluded), the octet cell and its pyramids, and the slabs on a patch of 112 octahedron and 32 tetrahedron pyramids.

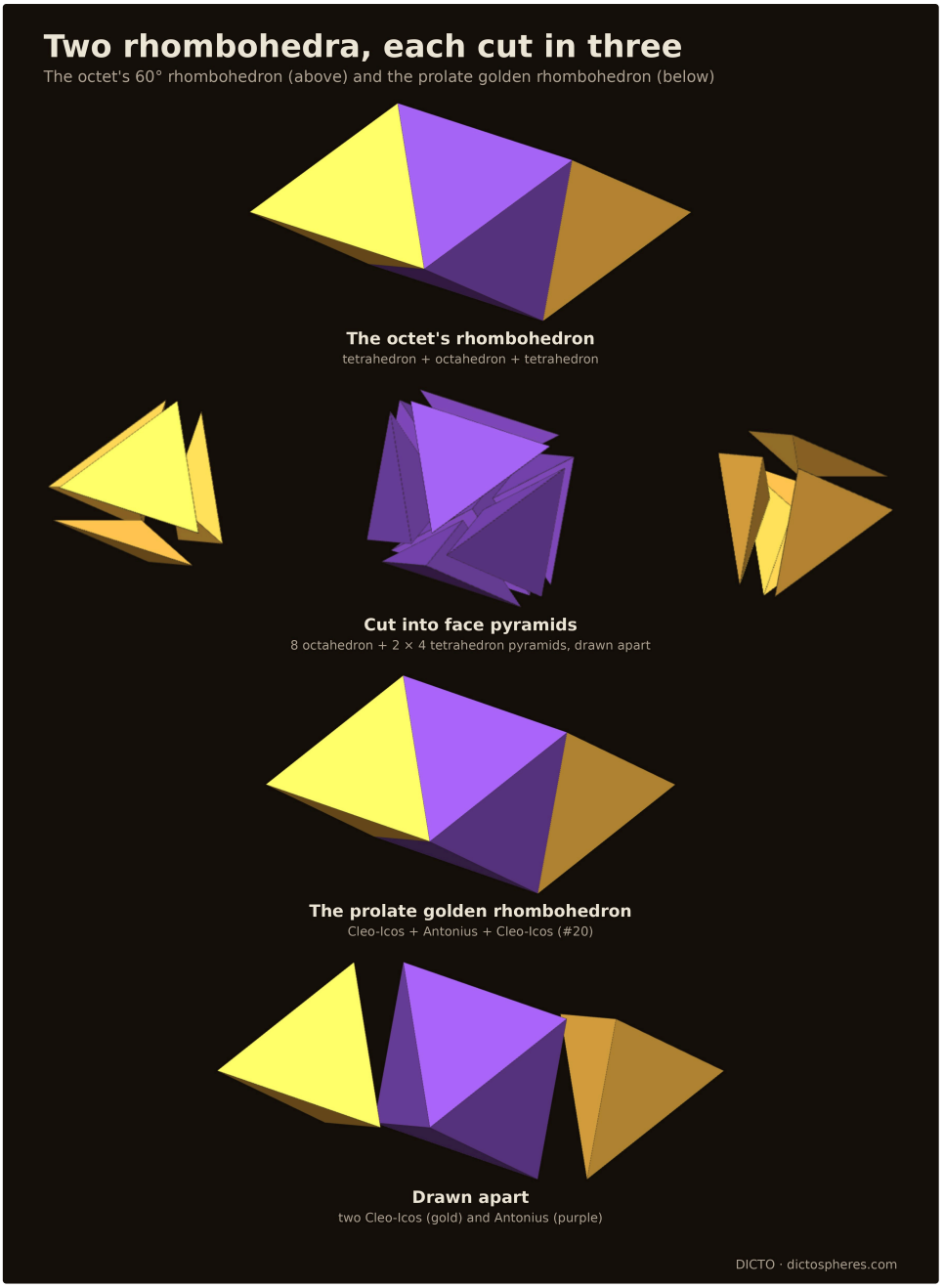


Figure 2. Above: one octet cell, the 60° rhombohedron of tetrahedron + octahedron + tetrahedron, whole (gold, purple, gold) and cut into face pyramids: the octahedron into its 8 pyramids (purple) and each tetrahedron into its 4 (gold), drawn slightly apart along their own axes. Below: the prolate golden rhombohedron, Cleo-Icos + Antonius + Cleo-Icos (#20), whole and drawn apart along its long axis. Each panel is scaled to fit.

3. Corollaries

> *Alone, only the cube's pyramid works; in pairs, only the tetrahedron's with the octahedron's.*

Corollary 1 (alone and in pairs). Alone, only the cube pyramid fills space. In pairs, only the tetrahedron and octahedron pyramids do; every other pair fails, the pairs with the cube pyramid because their second piece cannot be used.

> *When tetrahedron and octahedron pyramids fill space together, they come in equal numbers over large regions, as long as their edges are equal.*

Corollary 2 (equal numbers). In any tiling of space by tetrahedron pyramids of edge a and octahedron pyramids of edge b, the numbers $N_T(R)$ and $N_O(R)$ of each inside a ball of radius R satisfy $|a\,N_T(R) - b\,N_O(R)| \le C\,R^2$ for a constant C; so their numbers are in the ratio b : a in the limit, equal when the edges are equal, as in the octet.

Proof. With h(α) = 1, Lemma 1 gives a sum of 0 at all but finitely many points of each edge. Integrate along all edges inside the ball: Σ over tiles of Σ over their edges of h(angle) × (length inside the ball) = 0. A tetrahedron pyramid wholly inside contributes 3 × a × ½, an octahedron pyramid 3 × b × (−½); the tiles that meet the sphere are at most $C'R^2$ in number (bounded size, volume bounded below) and contribute a bounded amount each. ∎

> *The icosahedron's pyramid, the Cleo-Icos, can fill space only with a different kind of partner: DICTO's Antonius, two Cleo-Icos to one Antonius.*

Corollary 3 (the Cleo-Icos needs Antonius). The Cleo-Icos fills space with no set of face pyramids; a partner must be another kind of solid. Antonius is one [3, §5.7]: the three lines from the icosahedron's centre to the corners of a face meet at arccos(1/√5) ≈ 63.43°, the angle of the golden rhombus, so they span a prolate golden rhombohedron, and the planes x + y + z = 1 and 2 (in coordinates along those lines) cut it into a Cleo-Icos at each sharp corner and **Antonius** in the middle, a trigonal antiprism of exactly 4 times the Cleo-Icos's volume (the rhombohedron is 6 times). Golden rhombohedra tile space by a lattice, so **Cleo-Icos and Antonius fill space, two to one** (Figure 2, below). In the counting map's terms: along Antonius's 6 triangle edges the angle is $\pi - \theta_I/2 \approx 110.91°$, which with the Cleo-Icos's $\theta_I/2$ makes a flat 180°, and along each of its 6 zigzag side edges 108° (the rhombohedron's 72° edges are the 6 at its sharp tips, which belong to the Cleo-Icos); with $h(\theta_I) = 1$ and edges weighted by length, a Cleo-Icos carries $3 \times ½$ and Antonius $6 \times (-½)$, -2 times as much, which the two-to-one ratio balances (the Dehn invariant remark of [3]). The same cut of the 60° rhombohedron gives the octet's tetrahedron + octahedron + tetrahedron [3]: in both, the middle third is the partner. (Checked exactly in pyramid_checks.py.)

4. Conjectures

None: every statement above is proved. An open question, not conjectured here, is which tilings by tetrahedron and octahedron pyramids exist besides the octet cut up and its rearrangements.

5. The whole solids and their partners

> *The whole solids can all fill space as well, each with the right partners; the table gathers how.*

DICTO also proposed the question of how the whole Platonic solids fill space. His earlier exact work answers it for all five:

Solid	Fills space with	Source
cube	alone	classical
tetrahedron, octahedron	each other, two tetrahedra to one octahedron: the tetrahedral–octahedral honeycomb (Maurolico; Fuller's octet truss)	classical; Maurolico (1529) [1]
dodecahedron	Dogstars: dodecahedra on alternate cubes of a cubic lattice, each a cube with six of Euclid's roofs [5]; the hole in each other cube is the Dogstar; the dodecahedra take $(5 + \sqrt{5})/8 \approx 90.45\%$ of space	tiling and Dogstar: Hart (1996) [6]; exact: The DICTO Sunstar, #42 [7]
icosahedron	DICTO's icosahedral octet: an icosahedron + 6 edge roofs (Admiral Caps) is the regular octahedron, so icosahedra (edge 1) with their roofs in the octet's octahedra, and icosahedra of edge 1/2 with corner and edge caps in its tetrahedra, fill space; the icosahedra take $(175 - 75\sqrt{5})/12 \approx 60.79\%$	exact (released finding F21; not a separate paper) [8]; the icosahedron in the octahedron at the golden section is classical (Pacioli; Fuller's Jitterbug)

Table 2. How each whole Platonic solid fills space with partners. The idea of F21 is DICTO's: "the same principle as the dodecahedron and roofs, but with six" [8].

The pattern is the same for both golden solids. Euclid builds the dodecahedron on a cube with a roof on each face [5]; the roofs are what let dodecahedra sit on a cubic lattice, and what is left is the Dogstar. DICTO moved the roofs to the icosahedron: six edge roofs make it an octahedron, which the octet already places. The main theorem says why such help is needed: a solid's face pyramids carry half its dihedral angle along each base edge, and only for the cube ($\pi/2$) is that angle a rational multiple of π; the tetrahedron's and the octahedron's cancel each other (α and $\pi - \alpha$), while θ_D and θ_I cancel with nothing among the five.

6. Aristotle, briefly

> *What Aristotle wrote, how it was argued over for two thousand years, and where DICTO's question fits.*

"Among surfaces it is agreed that there are three figures which fill the place that contain them—the triangle, the square and the hexagon: among solids only two, the pyramid and the cube" (Aristotle, On the Heavens III.8, 306b, in Guthrie's translation as quoted by Lagarias and Zong [1]). Read as the regular tetrahedron, the claim is wrong: five tetrahedra round an edge leave 7.36° [1]. Lagarias and Zong trace the long correction [1]: Simplicius and Averroës held that twelve regular tetrahedra fill space round a point; others said twenty; Thomas Bradwardine observed that the icosahedron cuts from its centre into twenty congruent tetrahedra but could not decide whether they are regular (they are not: Lagarias and Zong note that their solid angle at the centre is about ten percent larger than a regular tetrahedron's); Paul of Middelburg asserted that no number of regular tetrahedra fills space round a point, arguing from Euclid's Book XIII; and Maurolico (1529) stated that cubes alone, and tetrahedra together with octahedra, fill space. Twenty regular tetrahedra can meet at a point with small gaps [1]: the near miss behind the commentators' twenty.

We do not claim to know what Aristotle meant; Lagarias and Zong take "pyramid" to be the regular tetrahedron [1], the usual reading. DICTO proposed that it may have been Bradwardine's tetrahedron, the Cleo-Icos. Read as a face pyramid, the answer is exact: Aristotle's two figures come out right only as the cube's own pyramid (trivially, since six make the cube) and as the tetrahedron's pyramid taken together with the octahedron's, Maurolico's correction [1] cut from the centres; the Cleo-Icos needs Antonius.

7. Conclusion

> *If Aristotle's pyramid is a face pyramid, his claim is now settled exactly; what he himself meant remains unknown.*

Read as a statement about face pyramids, Aristotle's claim is settled by the main theorem: the cube's pyramid fills space alone, and the only other way is the tetrahedron's and the octahedron's pyramids together, in asymptotically equal numbers when their edges are equal. That pair is the octet truss cut from its centres, the classical fix that Maurolico gave for the regular solids. The icosahedron's pyramid, the Cleo-Icos, cannot join any set of face pyramids; its golden fix is Antonius, with which it fills the prolate golden rhombohedron two to one (#20). For the whole solids the common thread is Euclid's roof: on the cube it makes the dodecahedron that fills space with Dogstars, and moved to the icosahedron as six edge roofs it makes the octahedron that the octet places. What stays open is which other tilings by tetrahedron and octahedron pyramids exist, and what Aristotle himself meant.

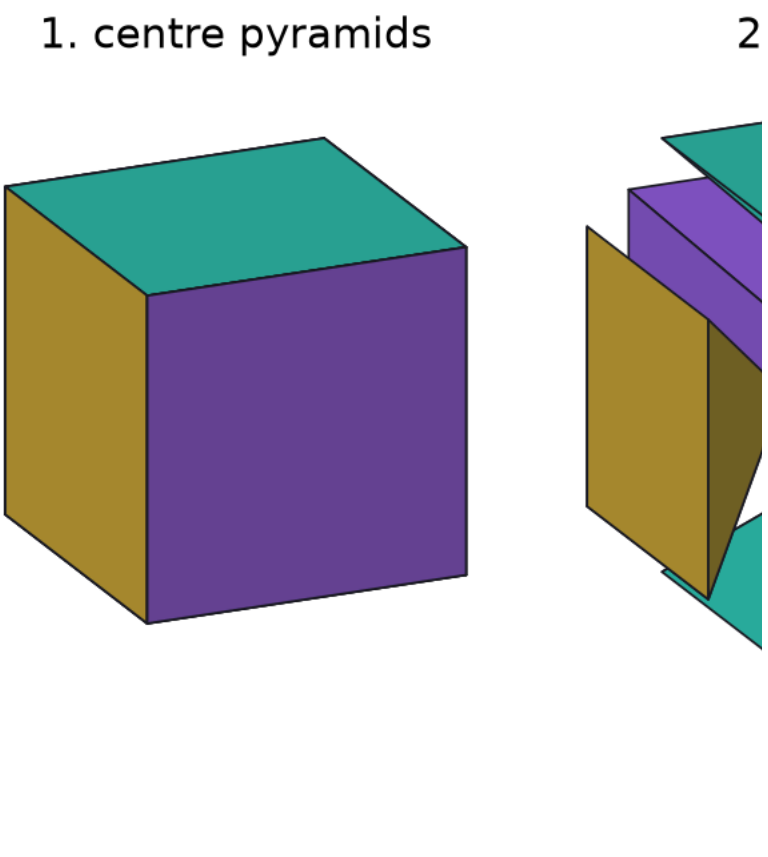


Figure 3. Part II of *The Centre Pyramids: from Plato to Kepler* (DICTO Vacuus Solids, #87) takes the same centre pyramids further, shown here for the cube: together they fill the cube (this paper); exploded; turned out into the Star, the rhombic dodecahedron of twice the volume; and emptied to the walls they share, the Vacuus, as sheets and as struts.

8. Prior art, and the status of each result

- **Known, credited:** the history of Aristotle's claim, the near miss, Bradwardine's remark on the icosahedron's twenty tetrahedra and Lagarias and Zong's note that they are not regular, Paul of Middelburg's argument from Euclid's Book XIII, Maurolico's tiling and the 60° rhombohedron of two tetrahedra and an octahedron (Lagarias and Zong [1]); Dehn's angle argument and the geodetic-angle theorems (Conway, Radin and Sadun [4]); the tiling of the cube's face pyramids (six make a cube); the dodecahedron–Dogstar tiling (Hart [6]); the icosahedron in the octahedron at the golden section (Pacioli; the Jitterbug); the Cleo-Icos itself, Mackay's unit tetrahedron and a pyramid of Hypsicles (as credited in [3]).
- **DICTO's earlier work, cited:** the Cleo-Icos and Antonius filling and the counting argument for the Cleo-Icos alone (#20 [3]); the exact Sunstar Lattice (#42 [7]); the icosahedral octet (F21 [8]).
- **Proved here in general:** the main theorem, with Lemmas 1 and 2, and Corollaries 1–3 (Corollary 3 with the filling of [3]). The principle behind it is classical: a polyhedron that tiles space has Dehn invariant zero (Debrunner [9]; Lagarias and Moews [10]); Lemma 1 with the counting map is a self-contained proof of the special case needed here, and what is new is the classification of the face-pyramid sets.
- **Proved for the cases built (exact checks):** the angles of Table 1, the octet cell and its 8 + 8 pyramids, the slabs that let cube pyramids join the octet, the golden rhombohedron's split and Antonius's angles.
- **Observed numerically:** only the PSLQ search of angle_relations.py, which finds no integer relation among π, α, θ_D, θ_I; it is evidence only, Lemma 2 being the proof.

In the sources checked (the full text of the 81 library sources with a text layer listed in the appendix, searched on 21 October 2026; the library's 175 link-only entries were not searched) we did not find a classification of the tilings by the face pyramids of the Platonic solids.

Data and reproducibility

In studies/2026-10-21-aristotle-pyramids: pyramid_checks.py (exact arithmetic in $\mathbb{Q}(\sqrt{5})$ with Python fractions; angles handled through $\cos^2$ and the sign of cos; log pyramid-checks.log, 47 PASS, no FAIL): the solids' dihedral angles; every base and side angle of the five pyramids; the facts used in Lemma 2; the test over all 31 sets; the octet cell and its 8 + 8 pyramids; the slabs of the octet patch; the golden rhombohedron, Cleo-Icos, Antonius and their angles. angle_relations.py (angle-relations.log): the angles to 80 digits and a PSLQ search finding no integer relation among π, α, θ_D, θ_I. Figures: aristotle_figs.py (float copies of the exact solids into the viewer of studies/2026-10-19-

five-fold-octet), jewel-shots.mjs (Playwright, 2× pixel ratio) and aristotle_compose.py. The 2,187-piece patch of Cleo-Icos and Antonius is that of [3].

References

1. J. C. Lagarias, C. Zong, Mysteries in packing regular tetrahedra, Notices Amer. Math. Soc. 59 (2012) 1540–1549, doi:10.1090/noti918 (quoting Aristotle, On the Heavens, tr. W. K. C. Guthrie, Loeb Classical Library 338, 1939, 306b).
2. DICTO, AXE, CLEO and FUJI, #19, DICTO DOI: 10.5281/zenodo.23351291.
3. DICTO, The FUJI Family, #20, DICTO DOI: 10.5281/zenodo.23357896 (§5.6 the Cleo-Icos, §5.7 Cleo-Icos and Antonius fill space).
4. J. H. Conway, C. Radin, L. Sadun, On angles whose squared trigonometric functions are rational, Discrete Comput. Geom. 22 (1999) 321–332; arXiv:math-ph/9812019 (Theorems 1–3).
5. Euclid, The Thirteen Books of the Elements, tr. T. L. Heath, vol. III (Books X–XIII), Cambridge (1908), Book XIII, Proposition 17.
6. G. W. Hart, Tetrahedral stellations of the dodecahedron (1996), https://www.georgehart.com/virtual-polyhedra/stellations-dodecahedron-tetrahedral.html (stellation 8).
7. DICTO, The DICTO Sunstar, #42, DICTO DOI: 10.5281/zenodo.23497702.
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9. H. E. Debrunner, Über Zerlegungsgleichheit von Pflasterpolyedern mit Würfeln, Arch. Math. 35 (1980) 583–587, doi:10.1007/BF01235384 (link only; not read).
10. J. C. Lagarias, D. Moews, Polytopes that fill $\mathbb{R}^n$ and scissors congruence, Discrete Comput. Geom. 13 (1995) 573–583, doi:10.1007/BF02574064; acknowledgment of priority, 14 (1995) 359–360, doi:10.1007/BF02570711 (link only; not read).

Appendix. The reference library searched

The library's manifests list 269 entries, of which 94 have a local file; these give the 81 distinct sources with a text layer listed below and 13 page images. The full text of the 81 sources was searched on 21 October 2026 for: Aristotle, De Caelo, On the Heavens, pyramid, fill space, space-filling, tile space, Dehn, dihedral angle, central pyramid, face pyramid, Maurolico, octet, tetrahedral-octahedral, golden rhombohedron, Cleo and Antonius. The relevant hits are Lagarias and Zong [1] (the history above), Conway, Radin and Sadun [4], the Dehn invariants of icosahedral tiles in Ogievetsky and Papadopolos, the tetrahedral-octahedral honeycomb (Wikipedia), and golden rhombohedra in Senechal, Dietl and Eschenburg and Kabai's demonstrations (all in the list below); none treats the face pyramids of the Platonic solids as tiles. The 13 page images without a text layer (the patents of Fuller, Snelson, Pearce, Baer, Hogan, Bergman and Lalvani, and Fedorov 1885) and the 175 link-only entries were not searched.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
E. S. Fedorov (chapter 13, English translation)	1885	Elements of the Theory of Figures, chapter 13
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
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T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
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M. Senechal	2004	Donald and the golden rhombohedra
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S. Kabai	2007-2015	Wolfram Demonstrations: 54-Faced Space-Filling Polyhedron (2007), Golden Octet Truss (2007), A Rhombic Dodecahedron Made of Rhombic Triacontahedra (2008), Relating a Rhombic Triacontahedron and a Rhombic Dodecahedron (2008), Cluster of 14 Rhombic Triacontahedra (2015)
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